

Hyperkalaemia 1: abnormal result

BMJ 2015;351:h4762, Clinical Practice Guidelines: treatment of acute hyperkalaemia in adults, UK kidney association 2023, Think kidneys: changes in kidney function and serum potassium during ACEi/ARB/diuretic treatment in primary care, October 2017, Derbyshire shared care pathology hyperkalaemia guideline, accessed 2020

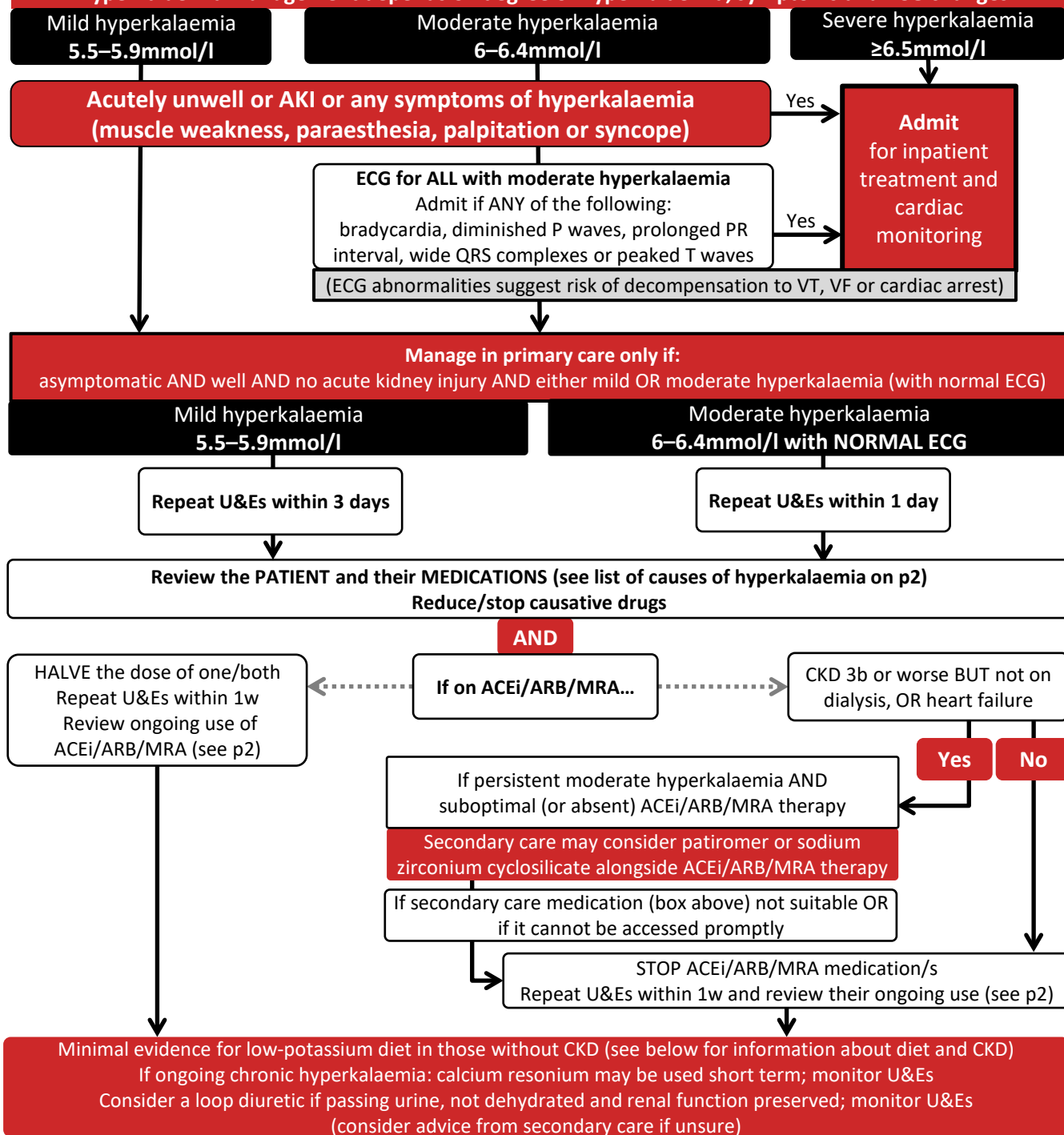
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Hyperkalaemia management depends on degree of hyperkalaemia, symptoms and ECG changes



CKD and hyperkalaemia

CKD and hyperkalaemia ≥5.5mmol/L:

- Assess for non-dietary causes, e.g. constipation, poorly-controlled diabetes.
 - Constipation reduces gut excretion of potassium (important in CKD because renal excretion is less efficient).
- If non-dietary causes corrected/not present: refer to a renal dietician.

CKD and sodium bicarbonate: if CKD and sodium bicarbonate <22mmol/L, sodium bicarbonate should be used with/without hyperkalaemia (likely a decision for the renal specialist!).

Drug acronyms

ACEi: angiotensin-converting enzyme inhibitor, e.g. ramipril

ARB: angiotensin receptor blocker, e.g. candesartan

MRA: mineralocorticoid receptor antagonist, e.g. spironolactone

Hyperkalaemia 2: causes and monitoring

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Hyperkalaemia: causes and risk factors

CAUSES:

- Drugs:
 - ACE inhibitors/ARBs/spironolactone/eplerenone.
 - Potassium supplements.
 - Potassium-sparing diuretic, e.g. amiloride (esp. with furosemide).
 - NSAIDs.
 - Macrogols, e.g. laxido, movicol, cosmocol.
 - Non-selective beta-blockers.
 - Trimethoprim/co-trimoxazole.
 - Digoxin toxicity.
 - Use of LoSalt in diet.
- Spurious result: delay processing sample, cold weather, sample-taking problems (small needle, clenched fist).
- Extreme physical exercise, crush injury or hypothermia (causes rhabdomyolysis).
- Excessive dietary intake, e.g. fruit, veg and processed foods (most significant if CKD).
- Tumour lysis syndrome.
- Addison's disease.

RISK FACTORS:

- Renal impairment, diabetes, heart failure, multimorbidity and/or serum bicarbonate <25mmol/L.

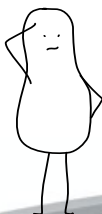
Community prevalence of hyperkalaemia is 0.5% (4.2% in CKD).

Prevention is better than cure

If CKD, heart failure or diabetes: monitor U&Es 2–4 times/year (increased risk of hyperkalaemia due to disease/drugs). Withhold ACEi/ARB/MRA during acute illness, and give information on sick day rules (risks of AKI and hyperkalaemia). Monitor U&Es more frequently if acute illness, previous AKI or previous hyperkalaemia. Caution when prescribing trimethoprim if on ACEi/ARB/MRA or renal impairment.

The hyperkalaemia see-saw

Hyperkalaemia is associated with an increased risk of hospitalisation, prolonged hospital stay and death.



The see-saw will be balanced differently for each individual.

Treatment of hyperkalaemia/initiation of sick day rules often requires a pause/discontinuation of a preventative drug. This risks suboptimal treatment of heart/renal disease and increases the risk of CV events, hospitalisation and death. In one study: mortality 4% if maintained on maximum ACEi/ARB/MRA, 8% if suboptimal dosing and 11% if treatment stopped.

Once potassium normalised

If ACE inhibitor/ARB/spironolactone/eplerenone has been stopped/halved, what was clinical indication for use?

- If for hypertension: consider alternative.
- If for heart failure with reduced ejection fraction OR kidney disease with albuminuria:
 - If had mild hyperkalaemia: continue but monitor carefully.
 - If had moderate hyperkalaemia: once potassium <5.5mmol/l, restart drug at lower dose (one at a time if >1 stopped) and monitor carefully.

Monitor for signs of heart failure, deterioration in renal function and/or rising potassium.

Starting and monitoring ACEi/ARB and MRAs

	ACEi/ARB	MRA
Before starting:	Check U&Es. Caution starting if K ⁺ >5.0mmol/L.	Check U&Es. Do not start if K ⁺ >5.0mmol/L or eGFR <30ml/min.
Monitoring on treatment:	Check U&Es 1–2w after starting and after each dose titration.	Check U&Es 1w after starting and after each dose titration; then monthly for 3m; then 3-monthly for the first year; then 4-monthly ongoing.

If creatinine rises, see *Acute kidney injury* article for management, including when to stop ACEi/ARB/MRA.

We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. July 2025. For full references see the relevant Red Whale articles.